

Supplementary Information: Unveiling Mycobacterium marinum Cellular and Metabolic Remodeling Under Simulated Microgravity

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1 Supplementary Tables Overview

All empirical data tables are archived in the repository under `results_tables/` and available via Zenodo:

- **Table S1** (`Table_S1_raw_counts_matrix.tsv`): Complete raw transcript count matrix (5,510 genes × 9 samples) generated by kallisto v0.52.0 pseudoalignment of NextSeq 550 paired-end reads.
- **Table S2** (`Table_S2_normalized_log2cpm_expression.tsv`): *Normalized* $\log_2(\text{CPM}+1)$ matrix across 4,964 expressed genes.
- **Table S3** (`Table_S3_sample_metadata.tsv`): Experimental metadata for NASA OSDR OSD-528.
- **Table S4** (`Table_S4_differential_expression_3dclinostat_v_s1g.tsv`): 351 significant DEGs in 3D Clinostat.
- **Table S5** (`Table_S5_differential_expression_rpm2_v_s1g.tsv`): 738 significant DEGs in RPM2.0 vs Static 1g.
- **Table S6** (`Table_S6_differential_expression_3dclinostat_v_s_rpm2.tsv`): 242 DEGs comparing simulation kin.
- **Table S7** (`Table_S7_wgcna_module_assignments_goslim.tsv`): WGCNA module assignments, standardized GO hub scores.
- **Table S8** (`Table_S8_wgcna_module_eigengenes.tsv`): Module Eigengene profiles across all 9 samples.
- **Table S9** (`Table_S9_wgcna_module_trait_correlations.tsv`): Pearson correlation coefficients and asymptotic p-values.
- **Table S10** (`Table_S10_goslim_pathway_enrichment.tsv`): GO Slim pathway over-representation metrics (kappa, p-value, FDR, Hochberg FDR).
- **Table S11** (`Table_S11_full_gene_ontology_enrichment.tsv`): Comprehensive GO enrichment table across Biological Processes.
- **Table S12** (`Table_S12_tabPFN_leave-one-out-cross-validation_predictions.tsv`): TabPFN leave-one-out cross-validation predictions.

- Table S13 (`Table_S13tabpfnfeatureiimportance.tsv`) : *TabPFN permutation feature importance rankings*
- Table S14 (`Table_S14cellularmetabolicmodelreactions.tsv`) : *Reaction-level GPR mapping across 32 enzymes*
- Table S15 (`Table_S15metabolicsubsystemperturbationiindex.tsv`) : *Quantitative Subsystem Perturbation Index*
- Table S16 (`Table_S16panmicrobialosdrcompendium78studies.tsv`) : *Compendium of 78 microbial space flight studies*
- Table S17 (`Table_S17cellularmetabolicmodelsbml.json`) : *SBML-compatible computational model schema*

2 Supplementary Figures Overview

All supplementary figures are archived as vector PDFs and high-resolution 300 DPI PNGs in `supplementary_figures/`:

- **Figure S1:** Systems architecture, biological model, simulation hardware, and analytical framework.
- **Figure S2:** Transcriptomic PCA biplot (70.1% PC1 variance) and volcano plot of 351 DEGs.
- **Figure S3:** Gene count distribution across 5 GOSlim co-expression modules and module-trait heatmap.
- **Figure S4:** TabPFN tabular foundation model benchmark under LOOCV, confusion matrix, and feature importances.
- **Figure S5:** Enriched GOSlim functional pathways with calibrated FAIR Blue-White-Red color fill.
- **Figure S6:** Intramodular hub connectivity (k_{within} vs k_{total}) and inter-module regulatory interactome.
- **Figure S7:** Pan-microbial spaceflight meta-analysis landscape across 78 OSDR studies and concordance matrix.
- **Figure S8:** Hexagonal trajectory radar mapping phenotypic concordance across 6 physiological axes.
- **Figure S9:** Multi-compartment cellular cross-section and Subsystem Perturbation Index (SPI) ranking.

3 Read Quality Control Pseudoalignment Convergence

Pseudoalignment against *M. marinum* M strain (NC_010612.1) converged with high pseudoalignment efficiency across all 9 biological replicates (mean 1.21 million reads mapped per sample). Normalized counts were converted to log-CPM values using edgeR TMM library scaling. Zero synthetic data were utilized in any calculation.